A SURVEY OF SOME RECENT DEVELOPMENTS
IN DIFFERENTIAL TOPOLOGY

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1. We consider differential topology to be the study of differenti-able manifolds and differentiable maps. Then, naturally, manifolds are considered equivalent if they are diffeomorphic, i.e., there exists a differentiable map from one to the other with a differentiable in-verse. For convenience, differentiable means C00; in the problems we consider, C' would serve as well.
The notions of differentiable manifold and diffeomorphism go back to Poincaré at least. In his well-known paper, Analysis situs [27] (see pp. 196-198), topology or analysis situs for Poincaré was the study of differentiable manifolds under the equivalence relation of diffeomorphism. Poincaré used the word homeomorphism to mean what is called today a diffeomorphism (of class C). Thus differential topology is just topology as Poincaré originally understood it.
Of course, the subject has developed considerably since Poincaré; Whitney and Pontrjagin making some of the major contributions prior to the last decade.
Slightly after Poincaré's definition of differentiable manifold, the study of manifolds from the combinatorial point of view was also initiated by Poincaré, and again this subject has been developing up to the present. Contributions here were made by Newman, Alexan-der, Lefschetz and J. H. C. Whitehead, among others.
What started these subjects? First, it is clear that differential geometry, analysis and physics prompted the early development of differential topology (it is this that explains our admitted bias toward differential topology, that it lies close to the main stream of mathe-matics). On the other hand, the combinatorial approach to manifolds was started because it was believed that these means would afford a useful attack on the differentiable case. For example, Lefschetz wrote [13, p. 361], that Poincaré tried to develop the subject on strictly "analytical" lines and after his Analysis situs, turned to com-binatorial methods because this approach failed for example in his duality theorem.
Naturally enough, mathematicians have been trying to relate these

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